



Early Exercise Exercises

1. Call options on stock

For a call option on stock to be an early exercise candidate, the following must hold true over the entire life of the option, as well as over the next day

$$\text{dividend} > \text{interest cost} + \text{volatility value}$$

The interest cost is the interest on the exercise price to expiration

The volatility value over the life of the option is approximately equal to the price of the companion out-of-the-money option.

The volatility value over the next day is approximately equal to the theta of the companion out-of-the-money option. Since this is almost always smaller than the dividend, the volatility value over the next day is usually not a consideration in early exercise of call options on stock.

If a stock pays no dividend over the life of the call option, there is never any reason to exercise the option early.



Early Exercise Exercises

1. Call options on stock

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

a.) Suppose there are only two choices, exercise now or hold the option to expiration. If the July 50 put is trading at a price of .20, which choice is best? Why?

b.) If this is a true American call, allowing for exercise at any time prior to expiration, when will you optimally decide whether to exercise the option?



Early Exercise Exercises

1. Call options on stock (continued)

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

- c.) If the stock price remains unchanged at 60.00 and the interest rate remains unchanged at 4.00%, on the day you must optimally decide whether to exercise the option, at what price must the July 50 put be trading in order for you to exercise the July 50 call?

- d.) If you decide to exercise the July 50 call but want the same protective characteristics offered by the July 50 call, what should you do?



Early Exercise Exercises

1. Call options on stock (continued)

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

- e.) If you take the appropriate action in question d., how much better off will you be with your new position than if you had not exercised the July 50 call?



Early Exercise Exercises

2. Put options on stock.

In the following questions ignore any short stock considerations.

For a put option on stock to be an early exercise candidate, the following must hold true over the entire life of the option, as well as over the next day

$$\text{interest earned} > \text{dividend} + \text{volatility value}$$

The interest cost is the interest on the exercise price to expiration

The volatility value over the life of the option is approximately equal to the price of the companion out-of-the-money option.

The volatility value over the next day is approximately equal to the theta of the companion out-of-the-money option.

The “blackout period” for a stock option put is the period prior to the payment of the dividend when it is not possible to earn enough interest through early exercise to offset the loss of the dividend.



Early Exercise Exercises

2. Put options on stock

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

- a.) What is the blackout period, in days, for this option? (Within what number of days prior to the dividend payment, would you never consider exercising the November 95 put?)

- b.) If there are 37 days until the dividend payment, at what price must the November 95 call be trading for you to consider immediate early exercise of the November 95 put?



Early Exercise Exercises

2. Put options on stock (continued)

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

c.) In addition to the price of the November 95 call, what must be the daily theta of the November 95 call for immediate exercise of the November 95 put to be optimal?

d.) If the daily theta of the November 95 call is .015 (in absolute value) would you exercise the November 95 put now?



Early Exercise Exercises

2. Put options on stock (continued)

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

- e.) If there are 10 days until the dividend payment, what must be the price of the November 95 call for you to want to immediately exercise the November 95 put?



Early Exercise Exercises (answers)

1. Call options on stock

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

- a.) Suppose there are only two choices, exercise now or hold the option to expiration. If the July 50 put is trading at a price of .20, which choice is best? Why?

$$\text{interest cost of exercising now} = 50 \times .04 \times 35/365 = .19$$

$$\text{interest cost + volatility value} = .19 + .20 = .39$$

You should exercise now in order to collect the dividend since $.39 < .50$

- b.) If this is a true American call, allowing for exercise at any time prior to expiration, when will you optimally decide whether to exercise the option?

In 14 days, since exercise of a stock option call can only be optimal the day before the stock pays the dividend.



Early Exercise Exercises (answers)

1. Call options on stock (continued)

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

- c.) If the stock price remains unchanged at 60.00 and the interest rate remains unchanged at 4.00%, on the day you must optimally decide whether to exercise the option, at what price must the July 50 put be trading in order for you to exercise the July 50 call?

Interest cost the day before the stock pays the dividend = $50 \times .04 \times 21/365 = .12$

The July 50 put must be trading at a price less than $.50 - .12 = .38$

- d.) If you decide to exercise the July 50 call but want the same protective characteristics offered by the July 50 call, what should you do?

Buy the July 50 put since: $\text{long put} + \text{long stock} = \text{long call}$



Early Exercise Exercises (answers)

1. Call options on stock (continued)

stock price = 60.00

time to July expiration = 35 days

interest rate = 4.00%

dividend = .50 payable in 15 days

You own a July 50 call which can be exercised early.

e.) If you take the appropriate action in question d., how much better off will you be with your new position than if you had not exercised the July 50 call?

You will improve your position by

.38 - the put price



Early Exercise Exercises (answers)

2. Put options on stock

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

- a.) What is the blackout period, in days, for this option? (Within what number of days prior to the dividend payment, would you never consider exercising the November 95 put?)

$$\text{interest which can be earned each day} = 95 \times .05 / 365 = .013$$

$$\text{blackout period} = .25 / .013 \approx 19 \text{ (days)}$$

- b.) If there are 37 days until the dividend payment, at what price must the November 95 call be trading for you to consider immediate early exercise of the November 95 put?

$$\text{interest which can be earned} = 95 \times .05 \times 54 / 365 = .70$$

$$.70 > \text{volatility value} + .25$$

The November 95 call must be trading at a price less than .45



Early Exercise Exercises (answers)

2. Put options on stock (continued)

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

- c.) In addition to the price of the November 95 call, what must be the daily theta of the November 95 call for immediate exercise of the November 95 put to be optimal?

One day's worth of interest value $.05 \times 95 / 365 = .013$

The theta must be less than .013 (in absolute value)

- d.) If the daily theta of the November 95 call is .015 (in absolute value) would you exercise the November 95 put now?

No. At a minimum, you should wait for the theta to fall below .013.



Early Exercise Exercises (answers)

2. Put options on stock (continued)

stock price = 82.50

time to November expiration = 54 days

interest rate = 5.00%

dividend = .25

You own a November 95 put which can be exercised early.

- e.) If there are 10 days until the dividend payment, what must be the price of the November 95 call for you to want to immediately exercise the November 95 put?

10 days is within the blackout period. You should never exercise during this period, regardless of the call price.